

# Wireless carrier waves

## 2.1 Basic principles and the physics involved

All electronic communication, may it be between humans or machines, wired or wireless, takes place through signals. A signal is a measurable physical quantity, the value of which varies with time in a meaningful manner. The simplest example of this can be found in a microphone, which converts sound (sound is really very small variations in air pressure) using a sensor to a varying electric voltage. Other properties of the signal are maintained. A larger pressure variation leads to a higher voltage change and a faster variation leads to a higher frequency signal. For further understanding of these concepts, let us understand the various properties associated with a signal.

### Amplitude

The amplitude of a signal is the maximum magnitude (size) of the varying parameter. In case of an electronic signal, the maximum voltage is the amplitude. In the case of an electromagnetic wave, the varying quantity is an electric field or a magnetic field.

### Frequency

The frequency of a signal is the number of times the signal varies in a second. The unit of frequency is hertz (Hz), named after the famous physicist of the same name. If the voltage varies once in a second (one variation is considered to be till

the voltage returns to the previous value. For example, if it goes from 0 to 5, and then back to -5 and back to 0. the frequency is considered to be 1 Hz. A thousand hertz is called a kilohertz (kHz), a million Hertz a megahertz (MHz), and so on.

**It was the discovery that the speed of radio waves and light is the same that led to the conclusion that light is an electromagnetic wave**